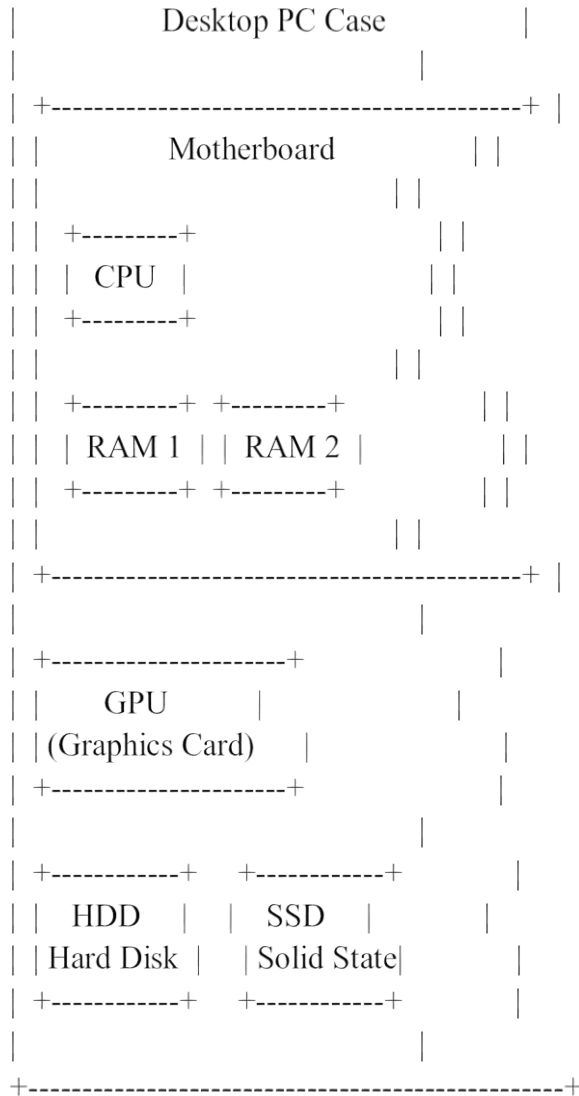


Hardware components theory

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



❖ Components:

- **Motherboard:** Main circuit board connecting all components.
- **CPU:** Central Processing Unit (brain of the computer).
- **RAM:** Temporary memory used by running programs.
- **GPU:** Processes graphics and gaming visuals.
- **HDD:** Traditional storage device with moving parts.
- **SSD:** Fast storage device with no moving parts.

2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.
Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower	Much Faster
Boot Time	30-60 seconds	10-20 seconds
Moving Parts	Yes	No
Durability	Less durable	More durable
Shock Resistance	Low	High
Noise	Produces noise	Silent
Power Consumption	Higher	Lower
Storage Cost	Cheaper per GB	More expensive per GB
Typical Use	Bulk storage, backups	Operating system, gaming, laptops
Gaming Performance	Longer loading times	Faster loading times

- SSD is faster and more reliable.
- HDD is cheaper and suitable for storing large amounts of data.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

❖ CPU (Central processing unit) :

- The CPU acts as the brain of the computer. In games like PUBG or Free Fire, it handles game logic, player actions, AI behavior, and communication between different hardware components.

❖ RAM (RANDOM ACCESS MEMORY) :

- RAM temporarily stores game data that the CPU needs quickly. More RAM allows the game to load maps, textures, and background processes smoothly without lag.

❖ GPU (graphic processing unit) :

- The GPU renders images, animations, shadows, and visual effects. A powerful GPU provides higher FPS (Frames Per Second), better graphics quality, and smoother gameplay.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Component	Why Upgrade?
GPU (Graphics Card)	FIFA uses detailed graphics and animations. A better GPU improves FPS and visual quality.
RAM	Upgrading to 16 GB or more helps the game run smoothly and reduces stuttering.
SSD	Faster game loading times and quicker system responsiveness.
CPU	A newer CPU improves game processing, AI calculations, and overall performance.
Power Supply (PSU)	Needed if upgrading to a more powerful GPU or CPU.
Cooling System	Prevents overheating during long gaming sessions.